

HARTNES STATE, VT, USA

WMO: 726115

Lat: 43.344N

Lon: 72.518W

Elev: 176

StdP: 99.23

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 54740

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-17.5	-26.3	0.3	-18.4	-23.7	0.5	-16.3	9.0	-4.0	8.2	-3.7	1.1	300	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	31.3	21.6	29.6	20.5	28.0	19.9	23.5	28.2	22.5	26.9	21.6	25.7	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.3	25.5	21.2	16.2	24.4	20.2	15.2	23.7	70.8	28.2	66.8	26.8	63.5	25.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.7	6.7	5.6	DB	-25.4	34.4	3.0	2.6	-27.6	36.3	-29.4	37.8	-31.1	39.2	-33.3	41.1	(4)
(5)			WB	-25.5	25.4	3.0	1.6	-27.7	26.6	-29.4	27.5	-31.1	28.4	-33.2	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-6.2	-4.6	-0.3	6.9	13.4	18.0	20.9	20.0	16.0	9.0	2.9	-2.6	(6)	
	DBStd	10.48	6.11	5.48	5.23	4.56	4.31	3.59	2.92	2.91	3.98	4.34	4.65	5.07	(7)	
	HDD10.0	2046	502	409	321	116	15	0	0	0	3	72	217	392	(8)	
	HDD18.3	4054	761	642	577	344	163	48	9	16	93	289	462	649	(9)	
	CDD10.0	1263	0	0	2	22	119	241	339	311	181	42	4	1	(10)	
	CDD18.3	229	0	0	0	1	9	39	89	68	21	1	0	0	(11)	
	CDH23.3	2280	0	0	5	32	167	416	859	597	192	11	0	0	(12)	
	CDH26.7	661	0	0	1	8	44	120	282	156	49	1	0	0	(13)	
(14) Wind	WSAvg	1.7	1.8	2.0	2.1	2.2	1.9	1.6	1.4	1.2	1.3	1.6	1.7	1.7	(14)	
(15) Precipitation	PrecAvg	1120	83	70	91	97	102	103	93	98	96	96	103	92	(15)	
	PrecMax	1430	204	231	210	208	286	311	184	225	213	314	184	207	(16)	
	PrecMin	770	11	5	17	18	26	17	25	21	8	9	29	36	(17)	
	PrecStd	163	47	35	39	41	50	60	44	45	48	61	40	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	11.9	21.2	27.3	30.3	32.5	33.8	32.3	31.1	24.9	18.3	12.8	(19)	
		MCWB	6.8	6.5	12.8	15.9	18.7	22.5	23.1	22.3	21.7	17.4	14.5	9.7	(20)	
	2%	DB	5.7	7.5	13.7	22.5	27.7	30.1	31.4	29.9	27.8	21.4	16.0	8.3	(21)	
		MCWB	3.2	3.4	8.0	12.9	17.5	20.9	21.9	21.1	20.1	15.7	12.3	5.8	(22)	
	5%	DB	3.1	4.9	9.9	18.9	24.8	27.7	29.8	28.4	25.3	18.7	12.7	6.1	(23)	
		MCWB	1.0	1.8	5.2	11.4	16.1	19.9	21.0	20.6	18.9	14.6	9.5	3.7	(24)	
	10%	DB	1.8	2.7	7.2	16.1	22.3	25.9	27.9	27.0	23.0	16.3	10.2	3.8	(25)	
		MCWB	0.1	0.3	3.8	9.4	14.8	18.3	20.1	19.7	17.8	12.7	7.3	1.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	6.9	13.9	17.6	20.8	24.9	24.8	24.4	23.4	19.5	16.2	10.4	(27)	
		MCDB	9.5	9.8	20.3	23.1	27.0	29.9	30.5	29.1	27.4	21.8	18.0	12.7	(28)	
	2%	WB	3.5	4.0	8.8	15.0	19.3	22.9	23.5	22.9	21.7	17.0	13.0	6.7	(29)	
		MCDB	5.3	6.7	12.1	20.9	25.0	27.7	28.5	27.0	25.4	19.9	15.1	8.2	(30)	
	5%	WB	1.6	2.1	5.9	12.4	17.8	21.4	22.5	21.9	20.4	15.3	10.2	4.1	(31)	
		MCDB	2.8	4.3	9.1	17.1	22.3	25.5	27.2	26.2	23.9	18.3	12.6	5.7	(32)	
	10%	WB	0.5	0.8	4.0	10.1	16.3	20.0	21.6	21.1	19.1	13.4	7.8	2.2	(33)	
		MCDB	1.5	2.1	7.0	15.4	20.6	23.8	26.0	24.6	22.3	15.7	9.5	3.5	(34)	
(35) Mean Daily Temperature Range	MDBR		9.4	10.3	10.4	12.2	12.8	12.1	12.4	12.3	12.4	11.1	9.3	8.1	(35)	
	5% DB	MCDBR	9.8	12.2	15.2	18.9	17.6	15.1	14.6	14.0	14.8	14.6	13.3	10.3	(36)	
		MCWBR	8.1	8.9	9.8	10.3	8.8	7.4	6.2	6.3	7.9	9.3	9.9	8.4	(37)	
	5% WB	MCDBR	9.2	10.9	13.8	16.3	14.2	12.8	12.4	11.6	12.1	12.6	11.9	9.6	(38)	
		MCWBR	7.9	8.4	9.7	10.2	7.8	7.2	6.0	6.1	7.4	8.9	9.7	8.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.294	0.321	0.355	0.387	0.427	0.446	0.420	0.370	0.338	0.308	0.284	(40)	
	taud		2.384	2.348	2.396	2.397	2.348	2.253	2.207	2.265	2.393	2.492	2.525	2.478	(41)	
	Ebn at Noon		845	904	921	911	886	847	825	836	857	846	824	823	(42)	
	Edh at Noon		78	95	104	113	122	135	140	127	104	83	68	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.61	2.49	3.65	4.50	5.03	5.42	5.64	5.09	4.13	2.56	1.73	1.24	(44)	
	RadStd		0.14	0.23	0.24	0.53	0.59	0.50	0.39	0.27	0.29	0.27	0.19	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	+33

Nomenclature: See separate page